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DEPARTMENT OF CSE

A.Y. 2026 – 2027
B.Tech. III Year Odd Semester

COURSE: CNAD (24CND3101)

PROJECT ABSTRACT SUBMISSION

TITLE OF THE PROJECT	Cloud-Native AI Research Discovery Platform	
S. No.	**University ID**	**Name**
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Abstract

The Cloud-Native AI Research Discovery Platform is a smart and scalable web-based application designed to address the difficulties faced by students, researchers, and academics in discovering, understanding, and organizing large amounts of research information. Traditional research requires searching multiple sources and reading lengthy papers, which can be time-consuming and inefficient. The proposed platform provides a centralized solution for searching relevant research papers, generating AI-based summaries, extracting keywords, identifying research insights, recommending related papers, and managing bookmarks and search history. The system uses React.js with Vite for the frontend, Node.js with Express.js for backend microservices, MongoDB for data management, Semantic Scholar API for research-paper discovery, and Google Gemini API for AI-powered analysis. Docker is used to containerize application services, while Kubernetes with Amazon EKS provides container orchestration, scalability, reliability, and cloud deployment on AWS. Git and GitHub support version control, and Postman is used for API testing. The expected outcome is an intelligent, modular, portable, and scalable research platform that reduces the time required to discover and understand academic literature while demonstrating practical cloud-native application development and AI integration.

Signature of the Guide **Signature of the Course Coordinator** **HOD**

Tools and Technologies Used for the Project:

Category	Tools / Technologies
Frontend	HTML5, CSS3, JavaScript, React.js, Vite
Backend	Node.js, Express.js
Database	MongoDB
Version Control	Git, GitHub
Containerization	Docker
Container Orchestration	Kubernetes / Minikube
Package Management	Helm
CI/CD	GitHub Actions
Security Scanning	Trivy
GitOps	Argo CD
Cloud Platform	AWS (Amazon EKS)
API Testing	Postman
Development IDE	Visual Studio Code
Monitoring	Prometheus, Grafana
Documentation	Github , Markdown

Signature of the Guide **Course Coordinator** **HOD**